1 Identity enrolment

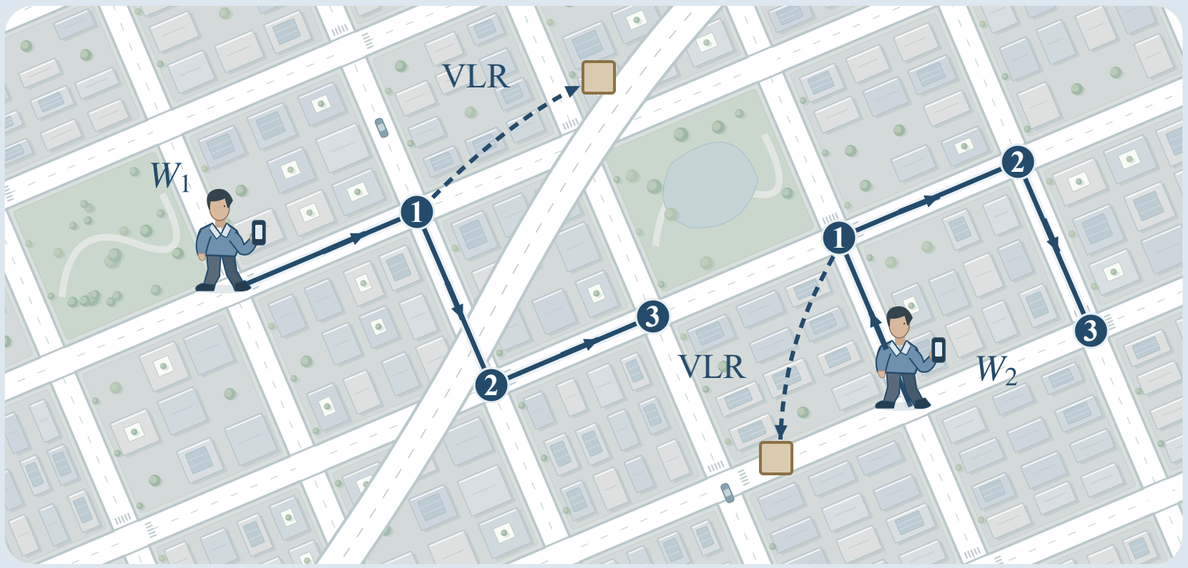
Sensing domain

Authority A

Unique participant credential

cert_A

Mobile sensing workers W



Solid: actual travel; dashed: VLR mapping to virtual cells.

Reports use real samples; gold squares are virtual cells.

2 Worker-local privacy

Signed participation; location / cost perturbation

Visited cells **Virtual swap**

$$\bar{b}_{i,t} = \text{VLR}(b_{i,t}) \longrightarrow \begin{bmatrix} p & 1-p \\ 1-p & p \end{bmatrix}_{\varepsilon_1} \longrightarrow \tilde{b}_{ij,t} \leftarrow \text{RR}_{\epsilon_1}(\bar{b}_{ij,t}), \quad p_\epsilon = \frac{e^\epsilon}{1+e^\epsilon}$$

Cost **Endpoint encoding**

Bounded input

$$z \sim \text{Bern}\left(\frac{c - c_{\min}}{c_{\max} - c_{\min}}\right) \longrightarrow \begin{bmatrix} p & 1-p \\ 1-p & p \end{bmatrix}_{\varepsilon_2} \longrightarrow \tilde{z}_{ij,t} \leftarrow \text{RR}_{\epsilon_2}(z_{ij,t})$$

Bit-gated cost release

$$\tilde{c}_{ij,t} = \tilde{b}_{ij,t}[c_{\min} + (c_{\max} - c_{\min})\tilde{z}_{ij,t}]$$

Public midpoint for an absent cell

Signed RR offer + signed reputation grade

$\tau_{i,t} = \text{Sign}_R(t, \text{sid}_{i,t}, \bar{r}_{i,t}, H(\text{tok}_{i,t}))$

$(\tilde{b}_{i,t}, \tilde{c}_{i,t})$

Broker S: validate + route

4 Selected reporting

Locked roster

Selected-worker receipt

L_t

$$\sigma_i = \text{Sign}_{a_{i,t}}(t, H(\text{ct}_{i,t}))$$

$$\text{Verify}_{a_{i,t}}(\sigma_i) = 1$$

Actual readings + region mask; fresh alias signature

Requester-only ciphertexts

Blockchain platform B

Bound inputs

Round binding $(t, \rho_t, \text{policy}, B_t)$

Old state → **Locked t** → **Next state**

Contract gates

- ✓ Roles / epoch
- ✓ Cap / roster
- ✓ State / replay

ρ_t Packets

$$C_i' = C_i s_i^{n_P}, \quad \text{sid}_{i,t+1} \leftarrow \{0,1\}^{\lambda_{\text{sid}}}$$

Open / Lock / Collect / Settle

Roles, bindings, cap and replay checks

6 State renewal

$\rho_t \longrightarrow \rho_{t+1}$

Broker S · single-round selection

3 Reputation-graded recruitment

Credible **Unknown**

$$u_t \in [u_{\min}, u_{\max}], \quad u_t \downarrow \text{ as } |C_t| \uparrow$$

Signed grades + perturbed virtual coverage

$$F(L) = \overline{v_j(L)} + \frac{\lambda_Q}{n} \sum_{i \in U_t} o_i(L)(a_0 + a_1 z_i)$$

$$i^* = \arg\max_i \frac{(\Delta F_i + \delta_F)(b_0 + b_1 \bar{r}_i)}{\max\{\Delta \mathbf{q}_i, \delta_q\}}$$

$$Q(L \cup \{i\}) \le B_t, \quad \bar{r}_i \in \Gamma_g$$

Prefer C/U; fall back only if absent or unaffordable

σ_t^+

Requester R · post-collection recovery and scoring

5 Credible anchors / secondary references / bounded update

$y^C_{\mathcal{A}_t}$ → **Kernel fit**

$$h_j = b_j + \mathbf{k}(y^C_{\mathcal{A}_t})^\top \alpha_j$$

$$\mathcal{H}_t = \{j \notin \mathcal{A}_t : \text{EV}_j \ge \zeta_R\}$$

Actual U report; no direct C reference

Task service

Decode

$$\hat{x}_j = a + \Delta_g \hat{g}_j$$

$$d_{ij} = \text{TD}(C_j \setminus \{i\}), \quad N_i = |D_i| + \beta |H_i|$$

$$r_i^+ = Q_h\left[\Pi_{[0,1]}\left(r_i + \sqrt{\min(N_i, K_E)}\, \bar{s}_i\right)\right]$$

Frozen references; no self-evaluation

Encrypted successors

$$\tau_i^+ = \text{Sign}_R(\text{hdr}_i^+)$$

Digest / serial / grade; ciphertext stays at S